

25-bit Vote Packing: k=10 candidates, N=255 bits



$$m = v_1 \cdot 2^{225} + v_2 \cdot 2^{200} + \dots + v_{10} \cdot 2^0$$

Constraint: $k \times b = 10 \times 25 = 250 \text{ bits} < \log_2(N) = 255 \text{ bits}$

Max per slot: $2^{25} - 1 = 33,554,431 \text{ votes}$

Homomorphism: Sum preserves slots independently: $\sum m_i = \sum_j (\sum_i v_{i,j}) \cdot 2^{\{25(k-j)\}}$